ANASTASIA S. LYULINA

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EDUCATION

Stanford University, *Stanford, CA, United States* 2020–*present*
PhD, Biology *(advised by Benjamin Good and Dmitri Petrov)*

Saint Petersburg Polytechnic University, *Saint Petersburg, Russia* 2013–2017
BSc, Physics

Technical University of Munich, *Munich, Germany* 2016

PROFESSIONAL APPOINTMENTS

Stanford University, *Stanford, CA, United States* 2020–*present*
Graduate Research Assistant, Departments of Biology and Applied Physics
ADVISORS: Benjamin Good, Dmitri Petrov

Institute of Science and Technology, *Maria Gugging, Austria* 2017–2020
Research Assistant, Evolutionary Genomics Group
ADVISOR: Fyodor Kondrashov

Harvard University, *Cambridge, MA, United States* 2019
Visiting Postgraduate Research Fellow, Systems Biology Department
ADVISOR: Johan Paulsson

University of Calgary, *Calgary, AB, Canada* 2017
Visiting Undergraduate Student, Centre for Molecular Simulation
ADVISOR: Peter Tieleman

Russian Academy of Sciences, *Saint Petersburg, Russia* 2016–2017
Undergraduate Student, Institute of Macromolecular Compounds
ADVISOR: Andrey Gurtovenko

FELLOWSHIPS AND AWARDS

Predoctoral Fellowship, Stanford Center for Computational, Evolutionary, and Human Genomics 2025–2026

Excellence in Teaching Award, Stanford Biology Department 2024

Sarah Ketterer Graduate Fellowship, Stanford University 2021–2024

Gruber Science Fellowship, Yale University *(declined)* 2020

Summer Research Internship, Zimin Foundation 2017

Academic Stipend, Saint Petersburg Polytechnic University 2015–2017

RESEARCH FUNDING

Mikitani Cancer Research Award, Stanford Cancer Institute *($40,000, with Saswati Karmakar)* 2024

PUBLICATIONS

** denotes equal contributions*

3. Karageorgi M, **Lyulina AS**, Bitter MC, Lappo E, Greenblum S, Mouza ZK, Tran CT, Huynh AV, Oken H, Schmidt P*, Petrov DA*. Beneficial reversal of dominance maintains a resistance polymorphism under fluctuating insecticide selection. *Nature Ecology and Evolution* 9, 2145–2160 (2025). *Highlighted by university press.*

2. **Lyulina AS***, Liu Z*, Good BH. Linkage equilibrium between rare mutations. *Genetics* 228, iyae145 (2024). *Featured on the cover.*

1. Gurtovenko AA, **Lyulina AS.** Electroporation of asymmetric phospholipid membranes. *Journal of Physical Chemistry B* 118, 9909–9918 (2014). *Featured on the cover.*

SUBMITTED MANUSCRIPTS

2. Konczal M, **Lyulina AS**, Zapata L, Camara F, Vlasova A, Saona R, Bello C, Fraïsse C, Gut M, Derelle R, Zheng C, Lin Y, Tutukina MN, Plyuscheva M, Fontsere C, Tomkovich PS, Iakushev NI, Shepelev IA, Clements J, Jarrett N, Clark N, Arkhipov VY, Zöckler C, Digby R, Lappo EG, Dong L, Li S, Hu J, Liu Y, Ossowski S, Marques-Bonet T, Guigó-Serra R, Syroechkovskiy EE, Kondrashov FA. Population growth facilitated the retention of deleterious variants in the critically endangered spoon-billed sandpiper *(submitted, accepted for publication).*

1. Karmakar S*, Xiong X*, **Lyulina AS***, Ozaki MK, Hebert JD, Tang YJ, Andrejka L, Lopez SS, Wang J, Charville GW, Flowers B, Stanger B, Tuveson DA, Batra SK, Tessier-Lavigne M, Good BH, Attardi LD, Petrov DA, Winslow MM. Context-specific regulatory rewiring dictates pancreatic ductal adenocarcinoma metastatic fitness *(submitted, under review).*

MANUSCRIPTS IN PREPARATION

2. **Lyulina AS**, Rosen MJ, Severson AL, Juan J, Sebastian L, Tran V, Nie W, Lai I, Wall GD, Winters IP, Winslow MM, Good BH, Petrov DA. Dynamics of early cancer growth inferred from tracking tens of millions of clonal tumors *in vivo.*

1. **Lyulina AS**, Good BH. Genetic diversity under nonequilibrium demography.

SELECTED TALKS AND SEMINARS

Harvard Medical School, *Boston, MA, United States*	2026
Harvard University, *Cambridge, MA, United States*	2026
Cambridge University, *Cambridge, United Kingdom (online)*	2026
Santa Fe Institute, *Santa Fe, NM, United States*	2025
Evolution Meeting, *Athens, GA, United States*	2025
Society for Modeling and Theory in Population Biology Meeting, *Chicago, IL, United States*	2025
Stochastic Physics in Biology Gordon Research Conference, *Ventura, CA, United States*	2025
Society for Molecular Biology and Evolution Annual Meeting, *Puerto Vallarta, Mexico*	2024
American Physical Society March Meeting, *Las Vegas, NV, United States*	2023
Bay Area Population Genomics Conference, *Davis, CA, United States*	2023
Harvard Quantitative Biology Symposium, *Cambridge, MA, United States*	2019
Graduate University for Advanced Studies, *Hayama, Japan*	2018

ADDITIONAL TRAINING

Cold Spring Harbor Laboratory Yeast Genetics and Genomics Course 2022
Cold Spring Harbor, NY, United States

Okinawa Institute of Science and Technology Computational Neuroscience Course 2018
Onna, Japan

TEACHING AND MENTORSHIP

AT STANFORD UNIVERSITY:

Biology 192/387: Seminar on Statistical Methods in Ecology and Evolution, *guest lecturer* 2025

Biology 85: Evolution, *teaching assistant (taught by Dmitri Petrov and Molly Schumer)* 2024

Biology 114: Building up Developing Scientists, *guest lecturer* 2023

Biology 113/244: Fundamentals of Molecular Evolution, *teaching assistant (taught by Dmitri Petrov)* 2022

School of Molecular and Theoretical Biology, *course instructor, research mentor* 2014–2018

Saint Petersburg Physics Olympiad Summer Training Program, *course instructor* 2013–2015

MENTORSHIP: Supervised the research of two undergraduate students in Fyodor Kondrashov's group and multiple high school students through the School of Molecular and Theoretical Biology; mentored undergraduate and graduate students through mentorship programs at the School of Molecular and Theoretical Biology, Stanford Undergraduate Research Association, Stanford Women's Community Center, and Stanford Biology Department.

SERVICE AND OUTREACH

Stanford Biology PhD Preview Program, *volunteer* 2020–2026

Stanford Biology PhD Program Admissions Committee, *student representative* 2023–2024

Stanford Biology Department Mentorship Committee, *committee member* 2021–2022

School of Molecular and Theoretical Biology Alumni Association, *board member* 2015–2017

PEER REVIEW: *Genetics*, *PLOS Genetics*, *PLOS Computational Biology*, *Journal of Theoretical Biology*, and (jointly with advisor) *PNAS*.

REFEREES

Benjamin Good, Assistant Professor of Applied Physics at Stanford University, *primary PhD advisor* bhgood@stanford.edu

Dmitri Petrov, Michelle and Kevin Douglas Professor of Biology at Stanford University, *PhD co-advisor* dpetrov@stanford.edu

Noah Rosenberg, Stanford Professor in Population Genetics and Society at Stanford University, *dissertation committee member* · noahr@stanford.edu

Monte Winslow, Associate Professor of Genetics and Pathology at Stanford University, *collaborator* mwinslow@stanford.edu